Pretraining Coverage Beats Scale: A Strong English–Dhivehi Model and Benchmark

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Abstract

Dhivehi is the national language of the Maldives. It is spoken by nearly half a million people and is written in Thaana, a right-to-left script that no other language uses. Despite this, none of the major open translation models support it: NLLB-200 does not include it, FLORES-200 has no Dhivehi data, and Google's recent TranslateGemma models do not list it among their benchmarked languages either. In this paper we describe how we trained a usable English→Dhivehi translation model with modest compute. We started from MADLAD-400 3B-MT, a T5-style encoder–decoder pretrained for translation across more than 450 languages, whose 256k-token vocabulary and pretraining corpus both cover Thaana; we confirmed this fit by measuring how efficiently different models' tokenizers encode Thaana text. We then fine-tuned it with LoRA on about 93,000 cleaned news-domain sentence pairs, roughly 10 GPU-hours of training. On a frozen test set of 1,500 sentence pairs, the chrF++ score improved from 42.6 (the model as downloaded) to **61.8**, and every output passed an automatic script-validity check that round-trips the text through the Segha keymap, a one-to-one mapping between Thaana characters and ASCII. We also compared against larger open models used out of the box (a quantized MADLAD 7B and the instruction-tuned LLMs TranslateGemma 12B, Gemma 4 12B and Qwen3 14B) and against earlier community fine-tunes for this direction; all of them scored below the fine-tuned 3B model. The strongest earlier system, an mT5-large trained on synthetic targets, trails it by 6 chrF++ on the same benchmark. The evaluation set construction steps, training code, and the LoRA adapter are released.

1 Introduction

Machine translation now covers hundreds of languages, but the coverage is uneven, and Dhivehi (ISO 639-1 `dv`) is a clear example of a language that has been left out. It is an Indo-Aryan language, the official language of the Maldives, and it is written in Thaana, a right-to-left script used by no other language. Because the speaker population is small, most large multilingual projects simply skip it. NLLB-200 [14] covers 202 languages, and Dhivehi is not one of them. Its companion benchmark FLORES-200 therefore has no Dhivehi portion, and neither did the earlier FLORES-101 [8]. TranslateGemma [7], Google's 2026 family of open translation models, reports results for 55 languages on WMT24++ [5], and Dhivehi is again absent. (Google Translate does support Dhivehi, but it is a closed commercial service, not something a researcher can inspect, run locally, or build on.) As far as we could find, there is no public benchmark for Dhivehi and no open translation model whose pretraining actually covered the language; what does exist publicly for English→Dhivehi is a handful of community fine-tunes [2, 13], and we evaluate all of them on our benchmark in §8.

The question this paper tries to answer is a practical one: how far can careful base-model selection and parameter-efficient fine-tuning close that gap? The main contributions are:

1. **A simple way to pick a base model for an unusual script.** Before doing any training, we measured subword fertility (the average number of tokens a tokenizer needs per word) on real Thaana text for several open models (§4). The differences were large: about a 6× spread between MADLAD-400's tokenizer and the tokenizers of recent general-purpose LLMs.

2. **A cleaned training corpus and a frozen benchmark** for English→Dhivehi, built only from permissively licensed public data (§5): 93,258 training pairs, a 1,000-pair development set, and a frozen, topic-stratified 1,500-pair test set.

3. **An automatic script-validity check.** Ev-

ery output is round-tripped through the Segha keymap [18], a one-to-one mapping between Thaana codepoints and ASCII characters. If an output contains anything that is not valid Thaana (plus allowed punctuation and digits), the round trip changes the string and the error is caught automatically (§2.2).

4. **Strong results from a well-matched base model.** LoRA fine-tuning of MADLAD-400 3B-MT for about 10 GPU-hours improved chrF++ by +19.2 points over the base model, and the result beats much larger out-of-the-box open models, including ones specialized for translation (§8).

5. **A reproducibility note** about a silent bug when loading MADLAD checkpoints under `transformers` 5.x. The bug ties the wrong weight matrices together and quietly turns the model's output into garbage (Appendix A).

2 Background

2.1 Dhivehi and Thaana

Dhivehi is spoken by nearly half a million people [6, 12], counting the resident population of the Maldives and the Mahl-speaking community on the Indian island of Minicoy. Thaana is written right-to-left. Every consonant letter carries an obligatory vowel diacritic (called *fili*), so well-formed text alternates between base characters (U+0780–U+07A5) and combining marks (U+07A6–U+07B0). Dhivehi morphology leans agglutinative, with long inflected verb forms; the news register in particular uses distinctive sentence-final perfective verb forms (for example, verbs ending in ކޮށްފި in headlines). Standard Dhivehi text uses Arabic-script punctuation (، ؛ ؟) rather than the ASCII equivalents.

2.2 The Segha keymap as a validity check

The Segha keymap [18] maps each Thaana codepoint to a single printable ASCII character and back. It was originally built for keyboard input and transliteration, but the fact that the mapping is one-to-one gives it a second use. For any string s that consists only of valid Thaana characters, common punctuation, digits, and whitespace, key2thaana(thaana2key(s)) = s. If the string contains anything else (mojibake, a stray Latin fragment, an Arabic-script character outside the allowed set, or an invalid codepoint), the round trip does not reproduce the input. We report the *round-trip pass rate*, the fraction of model outputs that survive this round trip unchanged, as a script-validity metric alongside the translation quality metrics. It is cheap and deterministic, and it catches a failure mode (mixing scripts) that chrF++ only penalizes weakly. The same module also supplies the punctuation normalization used during data preparation (mapping ASCII `,;?` to their Arabic counterparts).

3 Related work

MADLAD-400 [11] released a 3-trillion-token audited corpus covering 419 languages, along with T5-based [17] translation models trained on more than 450 languages. This matters here because they are, to our knowledge, the only strong open checkpoints whose training data genuinely includes Dhivehi (about 3.5M clean sentences). NLLB-200 [14] is the usual reference open system for low-resource translation, but it omits Dhivehi entirely. GATITOS [10] and SMOL [3] provide small professionally translated English–Dhivehi wordlists and sentences. On the data side, alakxender [1] has published aligned news-domain sentence pairs; their own MT models mostly target the dv→en direction, with en→dv, the direction needed to make content available *in* Dhivehi, covered only by a 250M multi-task Flan-T5 [2]. Neobe [13] released dedicated en→dv fine-tunes (mT5-large, ByT5-large, and a Qwen3-4B LoRA), trained on machine-translated (synthetic-target) news and Wikipedia text. NLLB-based Dhivehi fine-tunes also exist, but they target romanized Dhivehi (`div_Latn`) rather than Thaana. We evaluate the Thaana-native systems on our benchmark in §8. Finally, LoRA [9] and 4-bit NF4 quantization [4] keep the fine-tuning and the 7B-scale comparison runs affordable.

4 Choosing a base model with tokenizer fertility

A model is unlikely to translate well into a script that its tokenizer can only represent as individual bytes. To check this before training anything, we measured fertility (subword tokens per whitespace-separated word) on the 13,763 Dhivehi words in the frozen test set (Table 1).

MADLAD-400 and NLLB-200 tokenize Thaana compactly, while the general-purpose LLMs need 5–6× more tokens for the same text. That overhead hurts twice: the training targets become much longer (which spreads the learning signal thinner), and inference gets proportionally more expensive. Tokenizer coverage then combines with training-data coverage: MADLAD is the only model in the table that actually saw Dhivehi during pretraining. This pointed clearly at `madlad400-3b-mt`, a 3B-parameter T5-style

Tokenizer	tok/word	rel. to MADLAD
NLLB-200	2.78	0.96×
MADLAD-400	**2.89**	1.00×
Gemma-3	5.93	2.05×
Qwen3	16.05	5.55×
BLOOM	16.26	5.63×
OLMo-2	17.15	5.93×

Table 1: Subword fertility on the Thaana side of the test set. MADLAD's 256k SentencePiece vocabulary contains whole Thaana words, while most recent LLM tokenizers fall back to bytes. NLLB's tokenizer handles Thaana well, but the model itself was never trained on Dhivehi; tokenizer coverage alone is not enough.

encoder–decoder with a 256k SentencePiece vocabulary, trained for translation across more than 450 languages, which selects its target language with a task prefix (`<2dv> {English text}`). The encoder–decoder architecture also suits the task: the whole English sentence is encoded bidirectionally before any Thaana is generated, unlike in a decoder-only LLM. We picked the 3B size rather than 7B or 10B because it is the largest that still allows unquantized bf16 LoRA training within 16 GB.

5 Data

Sources. For this first round we used only the cleanest permissively licensed public data: (i) `dhivehi-english-translations` [1], 82,583 news-domain sentence pairs with topic labels (MIT license); and (ii) the GATITOS en–dv data [3, 10], 3,973 professionally translated words and short phrases (CC-BY-4.0), which we upsampled 3× so the model gets repeated exposure to reliable terminology. We deliberately left out the larger but noisier options (a 484k mixed-granularity crawl, and synthetic distillation sets); the synthetic sets in particular could contaminate any test set built from public news text.

Cleaning. Dhivehi text is normalized to Unicode NFC, and ASCII punctuation is mapped to its Arabic equivalents using the keymap module's rules. Pairs are dropped unless the target side is at least 70% Thaana characters, and pairs with a source/target length ratio outside $[0.25, 4.0]$ are filtered out.

Splits. From the source dataset's held-out test partition we froze two sets: *devtest* (1,500 pairs, topic-stratified, used only for final reporting) and *dev* (1,000 pairs, used for checkpoint selection). The training data was deduplicated against both splits on both language sides, leaving **93,258** training pairs.

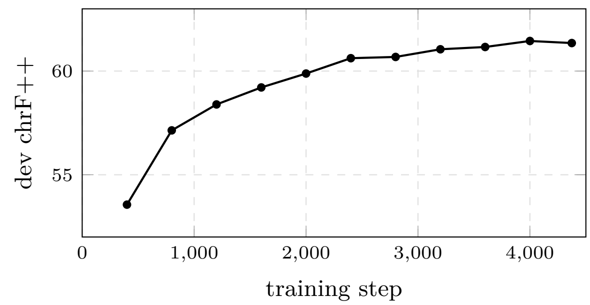

Figure 1: Development chrF++ (greedy decoding, 500-sample subset) during fine-tuning. The best checkpoint (step 4,000) is used for all reported results.

6 Fine-tuning setup

We trained LoRA [9] adapters (r=32, α=64, dropout 0.05) on all attention projections and feed-forward matrices of the encoder, decoder, and cross-attention (94.4M trainable parameters, or 3.1% of the model) in bf16, with gradient checkpointing and the 8-bit AdamW optimizer. The effective batch size is 64 (8 per device × 8 gradient-accumulation steps), the learning rate is 10^{-4} with cosine decay and 200 warmup steps, and training ran for 3 epochs (4,374 steps), about 10.3 hours end-to-end on our single 16 GB GPU. Every 400 steps we decoded a fixed 500-sample subset of the dev set greedily and kept the checkpoint with the best dev chrF++ (61.45, reached at step 4,000; Figure 1). Full hyperparameters are listed in Appendix C.

7 Evaluation protocol

All systems are scored on the frozen devtest set with sacreBLEU [16]. The headline metric is **chrF++** [15] (with word order 2), a character-level F-score that is better suited to a morphologically rich language like Dhivehi than word-level BLEU. As a secondary metric we report character-level BLEU (word-level BLEU is unreliable here because Thaana words carry so much inflection), plus the keymap round-trip pass rate from §2.2. The MADLAD-family systems decode with beam size 4. The instruction-tuned LLMs are evaluated zero-shot with a fixed prompt (Appendix B) at temperature 0. Because running them is slow, they are scored on a fixed random 400-pair subset of devtest (seed 13); the fine-tuned model and the 3B baseline are re-scored on that identical subset so the comparison is like-for-like.

System	chrF++	BLEU$_c$	RT
Full devtest (1,500), beam 4			
madlad400-3b-mt (base)	42.59	48.36	94.6%
madlad400-7b-mt-bt (NF4)	17.83†	8.78	80.3%
Flan-T5 en2dv 250M [2]	24.90	11.09	53.8%
Neobe ByT5-large 1.2B [13]	49.61	60.07	100%
Neobe mT5-large 1.2B [13]	55.75	67.15	100%
3B + LoRA (ours)	**61.82**	**72.21**	**100%**
400-pair subset, zero-shot instruction LLMs			
TranslateGemma 12B (q4)	33.18	40.83	96.3%
Gemma 4 12B (qat)	23.22	25.52	48.3%
Qwen3 14B (q4)	3.89	1.35	92.3%
madlad400-3b-mt (base)	42.19	47.49	93.3%
3B + LoRA (ours)	**62.77**	**73.16**	**100%**
NLLB-200	*no Dhivehi support*		

Table 2: English→Dhivehi results. BLEU$_c$ = character-level BLEU; RT = Segha keymap round-trip pass rate. Quantization is noted in parentheses. †Falls into repetition loops on about half of the inputs under NF4; §8 breaks this down.

8 Results

Fine-tuning improves the 3B model by **+19.2 chrF++** (42.6 → 61.8) and **+23.9** character BLEU, and closes the script-validity gap (94.6% → 100% round-trip). The base model occasionally emits stray Latin characters or malformed sequences; on this test set, the fine-tuned model never does (Table 2).

The instruction-LLM rows show two different ways a model can fail. Gemma 4 produces non-Thaana content in about half of its outputs (48.3% round-trip). Qwen3 is almost the opposite: its round-trip rate is high (92.3%) but its chrF++ is close to zero, because it tends to fall into repetition loops of valid Thaana words, and sometimes answers in the wrong script altogether (Sinhala or Arabic). In other words, script validity is necessary but not sufficient: the round-trip metric is meant to complement the quality metrics, not replace them. TranslateGemma, the strongest of the LLM rows, produces mostly valid Thaana but still scores below the *untuned* MADLAD 3B baseline, which is consistent with Dhivehi being outside its supported language set.

The quantized 7B row needs some unpacking. Under NF4, `madlad400-7b-mt-bt` falls into repetition loops on roughly half the test set (32% of its outputs contain no Thaana at all, and its average output is four times as long as the reference), which collapses its corpus-level score. On the 764 sentences where it decodes cleanly, it scores **49.0** chrF++ against 40.7 for the 3B base model on the same subset, so the larger model really is stronger when it stays stable. Quantization noise alone does not explain the failures: under identical NF4 settings the 3B model loses only 0.3 chrF++ (48.5 vs. 48.8 on a 150-sentence diagnostic), so the instability seems specific to this checkpoint under 4-bit quantization. Evaluating the 7B at full precision was outside our memory budget. The practical conclusion is unchanged: the fine-tuned 3B beats every alternative we could run.

Earlier en→dv fine-tunes. The multi-task Flan-T5 [2], evaluated with its author's recommended decoding settings, reaches 24.9 chrF++ with 53.8% round-trip validity. Neobe's dedicated models [13] are far stronger: on our benchmark their mT5-large sentence model reaches **55.75** chrF++ and the ByT5-large version 49.61, both with a perfect keymap round trip. Dedicated fine-tunes clearly get script validity right, and the mT5 is serious prior art. Our model is still ahead by **+6.1** chrF++, while training 13× fewer parameters (a 94M LoRA against a full 1.2B fine-tune) on human-aligned rather than synthetic targets. The margin is probably conservative, because Neobe's synthetic training pipeline draws on the same news domain as the test set. Overall, the pattern across Table 2 supports the premise of this paper: starting from a model whose pretraining covers the language beats both raw scale (the quantized 7B, the 12–14B LLMs) and task-only fine-tuning. None of the earlier systems report results on a shared public benchmark; this work closes that gap.

Qualitative behaviour. The outputs read as fluent news-register Dhivehi, including the correct headline perfective verb forms. Interestingly, the model often prefers native vocabulary where the reference uses an English loanword: for "Budget revised…" it produces އިސްލާހުކޮށް ("amended", a native word) where the reference has the loanword ރިވައިސްކޮށް. That is a defensible translation choice, but chrF++ counts it as an error. Table 3 shows examples. The remaining failure modes we observed are named-entity slips (one headline rendered *Urdu* with a form resembling "Erdoğan") and occasionally dropping honorific verb morphology.

9 Discussion and limitations

In-domain evaluation. The training and test data come from the same news distribution (though strictly deduplicated), so scores on conversational or technical text will be lower than the numbers reported here. **Single run.** All numbers come from one training run with one seed; we report no variance estimates. **Reference quality.** The test references are auto-aligned public data and contain occasional noise; we found cases of duplicated-sentence references where the model output was actually cleaner than the reference. If anything, then, the true quality is slightly under-measured. **Quantized comparisons.** The 7B

English source	Reference	Model output
Maldivian introduces premium economy tickets for domestic flights	މޯލްޑިވިއަން އިން އެތެރޭގެ އުދުހުންތަކަށް ޕްރީމިއަމް އިކޮނޮމީ ޓިކެޓް ތަޢާރަފުކުރަނީ	މޯލްޑިވިއަންގެ އެތެރޭގެ ދަތުރުތަކަށް ޕްރީމިއަމް އިކޮނޮމީ ޓިކެޓް ތަޢާރަފުކޮށްފި
Budget revised and increased to 17.95 billion rufiyaa	ބަޖެޓް ރިވައިސްކޮށް، 17.95 ބިލިއަން ރުފިޔާ އަށް ބޮޑުކޮށްފި	ބަޖެޓް އިސްލާހުކޮށް 17.95 ބިލިއަން ރުފިޔާއަށް ބޮޑުކޮށްފި
President's Office International Media Consultant Paul Roberts converts to Islam and marries Maldivian	ރައީސް އޮފީހުގެ އިންޓަރނޭޝަނަލް މީޑިއާ ކޮންސަލްޓަންޓް ޕޯލް ރޮބަޓްސް އިސްލާމްވެ ދިވެހިއަކާ ކައިވެނި ބައްލަވައިގެންފި	ރައީސް އޮފީހުގެ އިންޓަރނޭޝަނަލް މީޑިއާ ކޮންސަލްޓަންޓް ޕޯލް ރޮބަޓްސް އިސްލާމްވެ ދިވެހިއަކާ ކައިވެނިކޮށްފި

Table 3: Test-set examples (fine-tuned model, beam 4). In the last row the model's ކައިވެނިކޮށްފި is the plain register for "married"; the reference uses the honorific ކައިވެނި ބައްލަވައިގެންފި.

MADLAD row runs in NF4 and the LLM rows in 4-bit GGUF quantizations, so each of those rows slightly underestimates its full-precision counterpart; the observed gaps, however, are far larger than typical quantization losses. **No human evaluation yet.** Automatic metrics on a language with nearly half a million speakers deserve validation by native speakers, and we consider that the most important next step.

10 Conclusion

A carefully chosen pretrained translation model, about 93k cleaned public sentence pairs, and 10 GPU-hours of LoRA training were enough to build what is, as far as we can tell, the strongest open English→Dhivehi system available, ahead of much larger general and even translation-specialized open models, none of which cover Dhivehi. The overall recipe (screening base models by tokenizer fertility, starting from the cleanest data first, freezing a benchmark before training, and validating output script with a keymap round trip) should transfer to other excluded languages with unusual scripts. Future work: a second training round with the 484k-pair volume corpus, back-translation from 4.6M monolingual Dhivehi sentences, 7B-scale QLoRA, and native-speaker evaluation.

Artifacts. Training and evaluation code, the frozen benchmark, and this paper's source: `https://github.com/streetsolider/div-translation`. The fine-tuned LoRA adapter: `https://huggingface.co/str33t/madlad400-3b-mt-en-dv-news-v1`.

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A MADLAD loading bug in transformers 5.x

MADLAD checkpoints store the true shared embedding under `decoder.embed_tokens.weight` and a *separate* output projection under `lm_head.weight`, while their config declares `tie_word_embeddings=true`. `transformers` 4.x resolved this contradiction harmlessly, but some 5.x versions resolve it by tying *all three* matrices to the `lm_head` tensor, silently replacing the encoder/decoder embeddings and reducing generation to garbage. Our loader detects the bad tie (by checking whether `lm_head.weight == shared.weight`) and rewires both tensors from the checkpoint file, handling sharded checkpoints and quantized loads. Without this fix, the baseline numbers in Table 2 are not reproducible.

B LLM evaluation prompt

Zero-shot, temperature 0, one sentence per request:

```
Translate the following English sentence
into Dhivehi (the language of the
Maldives, written in Thaana script).
Reply with ONLY the Dhivehi translation
- no explanation, no romanization, no
quotes.
English: {sentence}
Dhivehi:
```

Responses are cleaned deterministically (thinking blocks and code fences stripped; the first Thaana-containing line is taken). Empty responses count against the model.

C Hyperparameters

Base model	`google/madlad400-3b-mt`
LoRA r / α / dropout	32 / 64 / 0.05
LoRA targets	q, k, v, o, wi_0, wi_1, wo
Trainable params	94.4M (3.1%)
Precision	bf16 (fp16 unstable for T5)
Optimizer	AdamW 8-bit
LR / schedule / warmup	10^{-4} / cosine / 200
Batch (device × accum)	$8 \times 8 = 64$
Max source/target length	256 / 256
Epochs / steps	3 / 4,374
Selection	best dev chrF++, eval every 400
Hardware	1× RTX 5070 Ti (16 GB), Windows
Wall-clock	∼10.3 h

Table 4: Fine-tuning configuration.